Fourie van Rooyen

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EDUCATION

The University of Alabama | Tuscaloosa, AL
Bachelor of Science in Aerospace Engineering | May 2026 | GPA: 3.0 | Dean's List Fall 2025
Relevant Coursework: Solid Mechanics, Aerospace Structures, High-Speed Aerodynamics, Propulsion, Thermodynamics

RELEVANT SKILLS

Programming/Software: SolidWorks CAD, CATIA, ANSYS (CFD simulations), MATLAB, Python, Delphi 10, Jira

Hardware/Technical: CNC Mill, CNC Lathe, 3D Printing, GD&T (ASME Y14.5-2018)

PROJECTS

Capstone Project, Tuscaloosa, AL, Chief Engineer — **August 2025 – May 2026**

- Led system-level design of a fixed-wing eVTOL tiltrotor reconnaissance UAV (Project ORCA) through a full NASA NPR 7123.1D design lifecycle, achieving stable VTOL hover and a successful VTOL→cruise→VTOL transition in flight testing.
- Executed full 3-D ANSYS Fluent CFD simulation at cruise ($V\infty = 17$ m/s), ANSYS Mechanical FEA (FoS ≥ 2.0 on primary structure), and transient thermal analysis; validated aerodynamic L/D ≈ 10.5–11.2 and zero structural yield under flight loads.
- Configured and flight-validated ArduPlane QuadPlane firmware on a Matek F405-Wing V2 flight controller with DSHOT600 ESCs and a PCA9685 I^2C tilt-servo expander; resolved TIM8 timer conflict enabling simultaneous DSHOT and PWM operation.
- Analyzed ArduPilot DataFlash flight logs to identify root causes of two in-flight anomalies (EKF3 altitude divergence, UV-induced PLA structural softening); produced a 38-page Post-Flight Analysis Report documenting anomalies, corrective actions, and lessons learned.
- Conducted experimental crush testing on 3D-printed structures, identifying 7% gyroid infill as optimal strength-to-weight configuration; oversaw cross-subsystem integration resolving CG, power, and mass tradeoffs through CDR.
- Oversaw cross-subsystem integration (aerodynamics, structures, propulsion, avionics, and manufacturing), resolving CG alignment, power, and mass tradeoffs across two full flight-test campaigns at a UA-approved outdoor UAV test area.

Alabama Aerodynamics Experimental Research Operations (A.A.E.R.O.)
Co-Founder / CFD and Testing Lead — **January 2025 – May 2026**

- Lead design of drone propeller with vortex inducers to evaluate aerodynamic performance.
- Simulated propeller thrust and acoustic characteristics using Computational Fluid dynamics (CFD) in ANSYS and SolidWorks and compared with real-world test data.
- Performed Finite Element Analysis (FEA) on designs to assess structural integrity and material performance.
- Applied Geometric Dimensioning and Tolerancing (GD&T) for precise manufacturing of components.
- Conducted static test stand experiments measuring thrust, RPM, and acoustics for multiple design iterations.
- Utilized wind tunnel for aerodynamic data collection and model validation.

Alabama Rocketry Association – Liquids Team Member — **October 2023 – May 2026**

- Collaborated on designing components for a single-stage, liquid propellant rocket, focusing on fluids systems.
- Designed test stand and created P&ID diagrams adhering to safety standards.
- Performed hydrostatic proof testing of flight hardware to demonstrate tank compliance with MIL-STD-810G/SMC-S-016 qualification requirements.
- Utilized a CNC Lathe to manufacture a graphite nozzle within design tolerances.
- Assisted in setup and execution of multiple hot fire tests for LOX/Kerosene engine, including short-duration and duration tests at Tuscaloosa National Airport, analyzing performance data to identify areas for improvement.

Genesis Drone Simulation Project, Personal Project — **August 2025 – May 2026**

- Built a high-fidelity quadcopter simulation in Python using the Genesis physics engine, modeling real-time aerodynamics, rigid-body dynamics, and gravity.
- Designed and tuned PID controllers for altitude, attitude, and position hold, achieving near-critical damping, sub-second settling times, and robust stability via anti-windup.
- Implemented manual and autonomous flight modes with differential motor RPM control and scaled the simulation to parallel multi-environment execution (>1000 steps/s on CUDA) with custom visualization tools.

WORK EXPERIENCE

The Cube, Tuscaloosa, AL, Lab Technician — **August 2025 – May 2026**

- Operate, maintain, and troubleshoot a fleet of advanced manufacturing equipment including FDM/FFF 3D printers, resin printers, waterjet cutting systems, and laser engravers to support academic, research, and student-led projects.
- Execute 3D printing using engineering-grade materials such as Nylon, Polycarbonate (PC), carbon-fiber–reinforced filaments, and specialty composites, optimizing print parameters for strength, surface finish, and dimensional accuracy.
- Support resin printing workflows, including part orientation, support strategy, washing, UV curing, and post-processing for high-resolution components.
- Assist students and faculty with CAD file preparation, slicing, considerations, ensuring manufacturability across additive and subtractive processes.